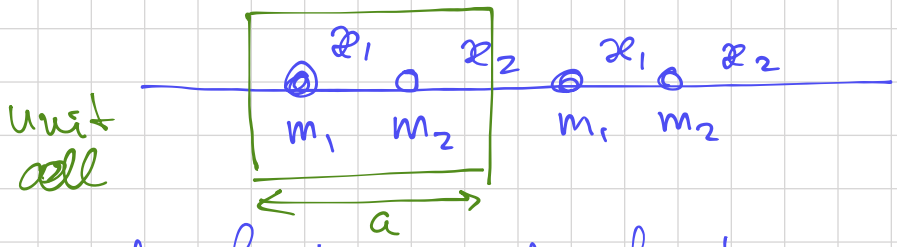
Phonon spectrum of a diatomic chain



Chain made of two kinds of atoms, m_1 and m_2 and spring constants $\varkappa_1, \varkappa_2$.

The unit cell — repeated unit of the lattice — contains m_1 and m_2.

Their positions are labeled as

$$x_1 = an + d_1 + \delta x_n$$

$$x_2 = an + d_2 + \delta y_n$$

deviations from equilibrium

positions inside the unit cell

unit cell # n

$$H = \sum_n \frac{p_{1n}^2}{2m_1} + \frac{p_{2n}^2}{2m_2} + \frac{1}{2}\varkappa_1 (\delta x_n - \delta y_n)^2 + \frac{1}{2}\varkappa_2 (\delta x_n - \delta y_{n-1})^2$$

$$\begin{cases} \delta \dot{x}_j = \dfrac{\partial H}{\partial p_{1j}} = \dfrac{p_{1j}}{m_1} \\ \delta \dot{p}_{1j} = -\dfrac{\partial H}{\partial x_j} = -\varkappa_1 (\delta x_j - \delta y_j) - \varkappa_2 (\delta x_j - \delta y_{j-1}) \end{cases}$$

$$\delta \ddot{x}_j = -\frac{\varkappa_1}{m_1}(\delta x_j - \delta y_j) - \frac{\varkappa_2}{m_1}(\delta x_j - \delta y_{j-1})$$

$$\begin{cases} \delta \dot{y}_j = \dfrac{\partial H}{\partial p_{2j}} = \dfrac{p_{2j}}{m_2} \\ \delta \dot{p}_{2j} = -\dfrac{\partial H}{\partial y_j} = \varkappa_1 (\delta x_j - \delta y_j) + \varkappa_2 (\delta x_{j+1} - \delta y_j) \end{cases}$$

$$\delta \ddot{y}_j = \frac{\varkappa_1}{m_2}(\delta x_j - \delta y_j) + \frac{\varkappa_2}{m_2}(\delta x_{j+1} - \delta y_j)$$

Fourier

$$\delta x_j = \frac{1}{\sqrt{N}} \sum_q e^{iqja} x_q$$

$$\delta y_j = \frac{1}{\sqrt{N}} \sum_q e^{iqja} y_q$$

number of unit cells

$$\frac{1}{\sqrt{N}} \sum_q e^{iqja} \ddot{x}_q = \frac{1}{\sqrt{N}} \sum_q e^{iqja} \left(\frac{\varkappa_1}{m_1}(y_q - x_q) + \frac{\varkappa_2}{m_1} \cdot \left(e^{-iqa} y_q - x_q\right)\right)$$

$$\frac{1}{\sqrt{N}} \sum_q e^{iqja} \ddot{y}_q = \frac{1}{\sqrt{N}} \sum_q e^{iqja} \left\{ \frac{\varkappa_1}{m_2}(x_q - y_q) + \frac{\varkappa_2}{m_2}\left(e^{iqa} x_q - y_q\right)\right\}$$

Harmonic motion $\Rightarrow \begin{cases} x_q(t) = e^{-i\omega t} A_x \\ y_q(t) = e^{-i\omega t} A_y \end{cases}$

$$\begin{cases} -\omega^2 A_x = \left(\frac{\varkappa_1}{m_1} + \frac{\varkappa_2}{m_1} e^{-iqa}\right) A_y - \left(\frac{\varkappa_1}{m_1} + \frac{\varkappa_2}{m_1}\right) A_x \\ -\omega^2 A_y = \left(\frac{\varkappa_1}{m_2} + \frac{\varkappa_2}{m_2} e^{iqa}\right) A_x - \left(\frac{\varkappa_1}{m_2} + \frac{\varkappa_2}{m_2}\right) A_y \end{cases}$$

Let's make masses equal, $m_1 = m_2$, as Simon does (only for simplicity).

$$m\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \varkappa_1 + \varkappa_2 & -\varkappa_1 - \varkappa_2 e^{-iqa} \\ -\varkappa_1 - \varkappa_2 e^{iqa} & \varkappa_1 + \varkappa_2 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

The eigenvalues are given by the roots of

$$\begin{vmatrix} \varkappa_1+\varkappa_2-m\omega^2 & -\varkappa_1-\varkappa_2 e^{-iqa} \\ -\varkappa_1-\varkappa_2 e^{iqa} & \varkappa_1+\varkappa_2-m\omega^2 \end{vmatrix} = 0$$

$$(\varkappa_1+\varkappa_2-m\omega^2)^2 = (\varkappa_1+\varkappa_2 e^{-iqa})(\varkappa_1+\varkappa_2 e^{iqa})$$

Two roots $\quad m\omega_\pm^2 = \varkappa_1+\varkappa_2 \pm \sqrt{\varkappa_1^2+\varkappa_2^2+2\varkappa_1\varkappa_2\cos(qa)}$

$$\omega_\pm = \sqrt{\frac{\varkappa_1+\varkappa_2}{m} \pm \frac{1}{m}\sqrt{(\varkappa_1+\varkappa_2)^2-4\varkappa_1\varkappa_2\sin^2\frac{qa}{2}}}$$

<u>Small-q behavior:</u> $\quad 4\varkappa_1\varkappa_2\sin^2\frac{qa}{2} \longrightarrow \varkappa_1\varkappa_2(qa)^2 \ll 1$

$\Rightarrow$ can expand $\sqrt{(\varkappa_1+\varkappa_2)^2-\varkappa_1\varkappa_2 q^2a^2} \overset{q\to 0}{\cong} \varkappa_1+\varkappa_2-\frac{\varkappa_1\varkappa_2}{2(\varkappa_1+\varkappa_2)}q^2a^2$

$$\Rightarrow \omega_\pm = \sqrt{\frac{\varkappa_1+\varkappa_2}{m} \pm \left(\frac{\varkappa_1+\varkappa_2}{m} - \frac{\varkappa_1\varkappa_2 a^2}{2m(\varkappa_1+\varkappa_2)}q^2\right)}$$

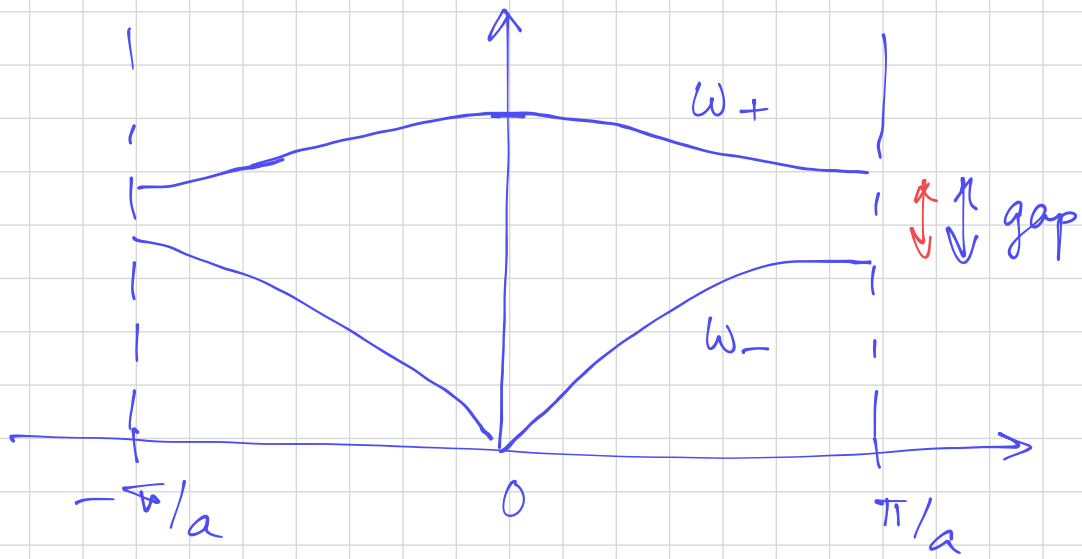
$$\omega_- = \sqrt{\frac{\varkappa_1\varkappa_2 a^2}{2m(\varkappa_1+\varkappa_2)}}\,|q| \quad \text{acoustic branch}$$

$$\omega_+ = \sqrt{\frac{2(\varkappa_1+\varkappa_2)}{m} - \frac{\varkappa_1\varkappa_2 a^2}{2m(\varkappa_1+\varkappa_2)}q^2} \quad \text{optical branch}$$

<u>Large-q:</u> $\quad qa = \pi \Rightarrow \sin^2\frac{qa}{2} = 1$

$$\omega_\pm = \sqrt{\frac{\varkappa_1+\varkappa_2}{m} \pm \frac{1}{m}\underbrace{\sqrt{(\varkappa_1+\varkappa_2)^2-4\varkappa_1\varkappa_2}}_{\sqrt{(\varkappa_1-\varkappa_2)^2}=\varkappa_1-\varkappa_2 \text{ (assume } \varkappa_1>\varkappa_2)}}$$

$$\omega_\pm = \sqrt{\frac{\varkappa_1+\varkappa_2}{m} \pm \frac{\varkappa_1-\varkappa_2}{m}} = \begin{cases} \sqrt{\frac{2\varkappa_1}{m}} & + \\ \sqrt{\frac{2\varkappa_2}{m}} & - \end{cases}$$



gap $\sqrt{\frac{2\varkappa_1}{m}} - \sqrt{\frac{2\varkappa_2}{m}} > 0.$

In this frequency interval waves do not propagate!

1 BZ : a = distance between unit cells, i.e. period of the lattice

Eigenvectors: $\underline{q \to 0}$

$$m\omega_\pm^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = (\varkappa_1+\varkappa_2) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

Acoustic mode: $\omega_-(q\to 0) \to 0 \Rightarrow \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} A_x \\ A_y \end{pmatrix} = 0 \Rightarrow \boxed{A_x = A_y}$

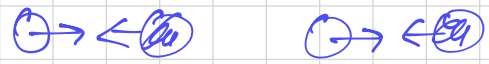
atoms 1 and 2 move <u>in-phase</u>

Optical mode: $\omega_+ \xrightarrow[q\to 0]{} \sqrt{\frac{2(\varkappa_1+\varkappa_2)}{m}}$

$$\Rightarrow \quad 2(\varkappa_1+\varkappa_2)\begin{pmatrix} A_x \\ A_y \end{pmatrix} = (\varkappa_1+\varkappa_2)\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \frac{1}{2}\begin{pmatrix} A_x - A_y \\ A_y - A_x \end{pmatrix}$$

$\Rightarrow$ now $A_x = -A_y \Rightarrow$ out-of-phase relative motion



Zone boundary: $qa = \pi \Rightarrow e^{\pm iqa} = -1.$

$$m\omega_\pm^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \varkappa_1+\varkappa_2 & \varkappa_2-\varkappa_1 \\ \varkappa_2-\varkappa_1 & \varkappa_1+\varkappa_2 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

$$m\omega_-^2 = m\,\frac{2\varkappa_2}{m} = 2\varkappa_2$$

$$\Rightarrow 2\varkappa_1 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \varkappa_1+\varkappa_2 & \varkappa_2-\varkappa_1 \\ \varkappa_2-\varkappa_1 & \varkappa_1+\varkappa_2 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

$$2\varkappa_2 A_x = (\varkappa_1+\varkappa_2)A_x + (\varkappa_2-\varkappa_1)A_y$$

$$-(\varkappa_1-\varkappa_2)A_x = -(\varkappa_1-\varkappa_2)A_y \Rightarrow \boxed{A_x = A_y} \text{ in-phase}$$

$$m\omega_+^2 = 2\varkappa_1 \Rightarrow$$

$$2\varkappa_1 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \varkappa_1+\varkappa_2 & \varkappa_2-\varkappa_1 \\ \varkappa_2-\varkappa_1 & \varkappa_1+\varkappa_2 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

so that $(\varkappa_1-\varkappa_2)A_x = (\varkappa_2-\varkappa_1)A_y$

$$\boxed{A_x = -A_y} \text{ out of phase}$$

Impurity atom (local defect)

$M \neq m$ (isotope)

$$H = \sum_n \frac{p_n^2}{2m}(1-\delta_{n,0}) + \frac{p_0^2}{2M} + \frac{1}{2}\varkappa(\delta x_n - \delta x_{n+1})^2$$

$$n \neq 0: \; m\delta\ddot{x}_n = \varkappa(\delta x_{n+1} + \delta x_{n-1} - 2\delta x_n)$$

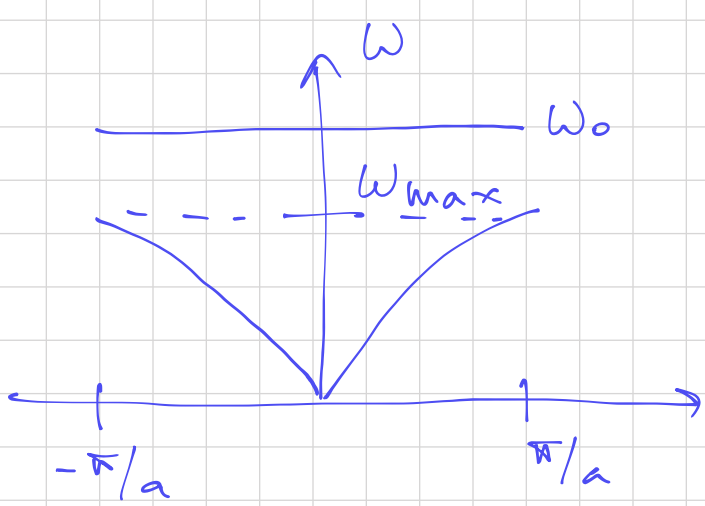
$$n = 0: \; M\delta\ddot{x}_0 = \varkappa(\delta x_1 + \delta x_{-1} - 2\delta x_0)$$

Impurity ($M \neq m$) disrupts propagating waves by scattering them, essentially it mixes waves with different q's.

It also creates new kind of oscillations — localized modes.

$$\delta x_n(t) \sim e^{-i\omega t}$$

$$\Rightarrow m\omega^2 \delta x_n + (M-m)\omega^2 \delta x_0 \underbrace{\delta_{n,0}}_{\text{Kronecker symbol}} = -\varkappa(\delta x_{n+1} + \delta x_{n-1} - 2\delta x_n)$$



Localized state must have $\omega_0 > \omega_{max}$

$$\omega(q) = 2\sqrt{\frac{\varkappa}{m}}\left|\sin\frac{qa}{2}\right|$$

(regular 1d chain)

$\omega(q)$ reaches maximum at $q = \frac{\pi}{a}$ $\Rightarrow$ to increase ω above $\omega(q=\pi/a)$ we need to consider <u>complex</u> $q = \frac{\pi}{a} + iK$

Then $e^{iqna} = \underbrace{e^{i\pi n}}_{(-1)^n} e^{-kna}$

Try $\delta x_n(t) = A(-1)^n \underbrace{e^{-k|n|a}}_{\text{must always choose decaying in space solution (not diverging)}} e^{-i\omega t}$

$\underline{n=0}$ $\quad M\omega^2 = -\varkappa\left(-e^{-ka} - e^{-ka} - 2\right) = 2\varkappa\left(1+e^{-ka}\right)$

$\underline{n\neq 0}$ $\quad m\omega^2(-1)^n = -\varkappa\left(-(-1)^n e^{-ka} - (-1)^n e^{ka} - 2(-1)^n\right)$

$$m\omega^2 = \varkappa\left(e^{\varkappa a} + e^{-ka} + 2\right)$$

$$\Rightarrow \quad e^{-ka} = \frac{M\omega^2}{2\varkappa} - 1$$

$$\frac{m\omega^2}{\varkappa} = 2 + \underbrace{\frac{M\omega^2}{2\varkappa} - 1}_{e^{-ka}} + \underbrace{\frac{1}{\frac{M\omega^2}{2\varkappa} - 1}}_{e^{ka}}$$

Let $\omega^2 = \frac{2\varkappa}{m}\lambda$ and $\frac{M}{m} = 1+y$, then

$$\Rightarrow \quad 2\lambda = 1 + \frac{2M}{2m}\lambda + \frac{1}{\frac{2M\lambda}{2m} - 1}$$

$$2\lambda = 1 + (1+y)\lambda + \frac{1}{(1+y)\lambda - 1}$$

$$\lambda = 1 + y\lambda + \frac{1}{(1+y)\lambda - 1}$$

$$(1-y)\lambda - 1 = \frac{1}{(1+y)\lambda - 1}$$

$$(1-y^2)\lambda^2 + 1 - (1-y)\lambda - (1+y)\lambda = 1$$

$$(1-y^2)\lambda = 2 \Rightarrow \lambda = \frac{2}{1-y^2}$$

Thus $e^{-ka} = \frac{M}{2\varkappa}\frac{2\varkappa\lambda}{m} - 1 = \frac{M}{m}\lambda - 1 = (1+y)\lambda - 1$

$$e^{-ka} = (1+y)\frac{2}{1-y^2} - 1 = \frac{2}{1-y} - 1 = \frac{1+y}{1-y}$$

$e^{-\varkappa a} < 1 \Longrightarrow$ only $y<0$ produces localized mode.

↳ this is requirement of decaying solution.

$$\boxed{\frac{M}{m} = 1+y \Rightarrow y<0 \text{ means } M<m. \quad \omega_{loc}^2 = \frac{4\varkappa}{M\left(2-\frac{M}{m}\right)}}$$

$$\omega^2 = \frac{2\varkappa}{m}\lambda = \frac{4\varkappa}{m(1-y^2)}$$

$$e^{-ka} = \frac{1+y}{1-y} \Rightarrow (1-y)e^{-ka} = 1+y$$

$$e^{-ka} - 1 = y(1+e^{-ka})$$

$$y = -\frac{1-e^{-ka}}{1+e^{-ka}}$$

So that $1-y^2 = 1 - \left(\frac{1-e^{-ka}}{1+e^{-ka}}\right)^2 = \frac{1+2e^{-ka}+e^{-2ka}-(1-2e^{-ka}+e^{-2ka})}{(1+e^{-ka})^2}$

$$= \frac{4e^{-ka}}{\left[e^{-ka/2}\left(e^{ka/2}+e^{-ka/2}\right)\right]^2} = \frac{1}{\cosh^2\left(\frac{ka}{2}\right)}$$

So that $\omega^2 = \frac{4\varkappa}{m}\operatorname{ch}^2\left(\frac{ka}{2}\right)$, $\underbrace{\omega_{loc} = 2\sqrt{\frac{\varkappa}{m}}\operatorname{ch}\frac{ka}{2}}_{\text{dispersion relation for the frequency of localized mode.}}$.

Let us compare with propagating mode:

$$\omega(q) = 2\sqrt{\frac{\varkappa}{m}}\left|\sin\frac{qa}{2}\right|$$

Localized mode has $q = \frac{\pi}{a} + i\kappa$

$$\sin\frac{qa}{2} = \sin\left(\frac{\pi}{2} + \frac{i\kappa a}{2}\right) = \cos\left(\frac{i\kappa a}{2}\right) = \frac{1}{2}\left(e^{i\frac{i\kappa a}{2}} + e^{-i\frac{i\kappa a}{2}}\right) =$$

$$= \frac{1}{2}\left(e^{-\kappa a/2} + e^{\kappa a/2}\right) = ch\left(\frac{\kappa a}{2}\right).$$

Therefore $\omega_{loc}(\kappa) = \omega\left(q = \frac{\pi}{a} + i\kappa\right)$

* Frequency of localized mode is obtained by <u>analytic continuation</u> of phonon dispersion $\omega(q)$ to the complex wave vector $q = \frac{\pi}{a} + i\kappa$, with real κ.